HOUSE PRICE PREDICTION

Regression Concepts & Final Project Notes

Easy English + Hinglish explanations | Definitions, formulas, model comparison, Gradient Descent, evaluation metrics, and business interpretation

Project goal: Predict a house's price using features such as area, bedrooms, bathrooms, location score, distance from the city, lot size, garage, pool, and renovation information.

Target variable: `house_price_inr`
Important note: `house_id` is an identifier, so it is normally excluded from model features.

Study Notes • Definitions • Key Formulas • Interpretation

1. Regression Basics

1.1 Regression

Definition: Regression is a supervised machine-learning technique used to predict a continuous numerical value, such as house price, salary, or sales.

Easy explanation / use: In this project, the model predicts house prices in INR from property features.

1.2 Feature and Target

Definition: A feature is an input column used for prediction. The target is the value the model tries to predict.

Easy explanation / use: Examples: area_sqft and bedrooms are features; house_price_inr is the target.

1.3 Training Data and Testing Data

Definition: Training data is used to fit the model; testing data is held back to evaluate how well the model predicts unseen examples.

Easy explanation / use: An 80/20 split means about 80% for training and 20% for testing. The test set should not be used to fit the model.

1.4 Feature Scaling

Definition: Feature scaling transforms numerical columns to comparable scales so that features with large numeric values do not dominate optimization.

Easy explanation / use: Standardization is particularly useful for Gradient Descent. Fit the scaler on training data only, then transform both train and test data.

1.5 Standardization

Definition: Standardization subtracts the training mean and divides by the training standard deviation.

Formula:

```
z = (x - mean) / standard deviation
```

Easy explanation / use: After scaling, values are typically centered around 0 with standard deviation near 1.

2. Regression Models

2.1 Simple Linear Regression

Definition: A regression model that uses one input feature to predict a continuous target.

Formula:

```
y_hat = b0 + b1*x
```

Easy explanation / use: Your project uses area_sqft to predict house_price_inr. It is easy to interpret, but one feature may not capture all price factors.

2.2 Multiple Linear Regression

Definition: A regression model that uses two or more input features to predict the target.

Formula:

```
y_hat = b0 + b1*x1 + b2*x2 + ... + bp*xp
```

Easy explanation / use: Your model uses several property features together. In your results it performed substantially better than the one-feature model.

2.3 Polynomial Regression

Definition: A regression approach that adds powers of a feature, such as x^2, so the model can represent a curved relationship.

Formula:

```
y_hat = b0 + b1*x + b2*x^2
```

Easy explanation / use: Your degree-2 model uses area_sqft and area_sqft squared. Its test performance was almost the same as simple linear regression, so the extra curve added little benefit.

2.4 Linear vs Polynomial Plot

Definition: A plot that shows observed data points and fitted linear and polynomial prediction curves.

Easy explanation / use: If the two curves nearly overlap, the polynomial term may not add much predictive value. A scatter of points around either curve also shows natural variation and prediction error.

2.5 Actual vs Predicted Plot

Definition: A scatter plot comparing actual target values with the model's predicted values.

Easy explanation / use: Points close to the diagonal y = x line indicate predictions close to actual values. Points far from the line indicate larger errors.

3. Model Evaluation Metrics

3.1 Mean Absolute Error (MAE)

Definition: The average absolute difference between actual and predicted values. It is expressed in the target's units, INR in this project.

Formula:

```
MAE = mean(|y - y_hat|)
```

Easy explanation / use: Lower MAE is better and is often easy to explain as the average size of an error.

3.2 Mean Squared Error (MSE)

Definition: The average of squared prediction errors. Squaring gives greater weight to large errors.

Formula:

```
MSE = mean((y - y_hat)^2)
```

Easy explanation / use: Lower MSE is better, but its unit is INR squared, so it is less intuitive to explain to a business audience.

3.3 Root Mean Squared Error (RMSE)

Definition: The square root of MSE. It is in the same units as the target and penalizes large errors more than MAE.

Formula:

```
RMSE = sqrt(mean((y - y_hat)^2))
```

Easy explanation / use: Lower RMSE is better. In this project it is reported in INR.

3.4 R-squared (R^2)

Definition: A metric describing how much variation in the target is explained by the model relative to a mean-based baseline.

Formula:

```
R² = 1 - (SS_res / SS_tot)
```

Easy explanation / use: A higher value generally indicates better fit on the evaluated dataset. R^2 is not the same as prediction accuracy and does not guarantee good performance on future data.

3.5 Adjusted R-squared

Definition: A version of R^2 that adjusts for the number of predictors and the sample size.

Formula:

```
Adjusted R² = 1 - (1 - R²) * (n - 1) / (n - p - 1)
```

Easy explanation / use: It can help compare models with different numbers of features. It may decrease when added predictors do not provide enough useful information.

Symbols: y = actual value; y_hat = predicted value; n = number of evaluated observations; p = number of predictors (excluding the intercept).

4. Gradient Descent Optimization

4.1 Gradient Descent

Definition: An optimization method that repeatedly updates model parameters in the direction that reduces the loss function.

Formula:

```
theta_new = theta_old - learning_rate * gradient
```

Easy explanation / use: It is an optimizer, not a separate regression model. The learning rate controls the size of each update.

4.2 Batch Gradient Descent

Definition: Calculates the gradient using the full training dataset before each parameter update.

Easy explanation / use: Its updates can be stable, but each update can be computationally expensive on large datasets.

4.3 Stochastic Gradient Descent (SGD)

Definition: Updates parameters using one training example at a time.

Easy explanation / use: It can make many updates quickly, but its loss curve is often noisy because each update uses only one sample.

4.4 Mini-Batch Gradient Descent

Definition: Updates parameters using a small batch of training examples.

Easy explanation / use: It balances the full-data update of Batch GD with the frequent updates of SGD. A batch size such as 32 is a common starting point.

4.5 Learning Rate and Epoch

Definition: The learning rate controls update size; an epoch means one complete pass through the training dataset.

Easy explanation / use: A very small learning rate can make learning slow. A very large rate can cause unstable updates or prevent convergence.

4.6 Loss vs Epoch Plot

Definition: A convergence plot showing training loss across epochs for each optimizer.

Easy explanation / use: A downward trend suggests learning is progressing. Compare methods using clearly stated settings; equal epochs do not mean equal numbers of parameter updates.

5. Bias, Variance, Underfitting and Overfitting

5.1 Bias

Definition: Bias is error caused by a model's simplifying assumptions. A model with too much bias may fail to capture important patterns.

Easy explanation / use: A one-feature model may have higher bias if house prices depend on several property characteristics.

5.2 Variance

Definition: Variance describes how sensitive a model is to changes in its training data.

Easy explanation / use: Very flexible models may learn sample-specific patterns that do not generalize well.

5.3 Underfitting

Definition: Underfitting occurs when a model is too simple to learn the important patterns in the data.

Easy explanation / use: Typical evidence: both training and test performance are poor.

5.4 Overfitting

Definition: Overfitting occurs when a model fits training data very closely, including noise, but performs worse on unseen data.

Easy explanation / use: Typical evidence: training performance is much better than test performance. One test score alone cannot confirm overfitting.

5.5 Bias–Variance Trade-off

Definition: The trade-off is the balance between a model that is too simple and one that is too sensitive to training data.

Easy explanation / use: Choose a model that performs well on unseen data, ideally confirmed with cross-validation, rather than selecting only by training score.

6. Your Project Results and Interpretation

The following values are based on the results discussed while working on this project. They should be checked against the final notebook output before submission.

Model	Test R^2 (approx.)	Test RMSE (approx.)	Interpretation
Simple Linear Regression	0.5625	■81.85 lakh	Uses area_sqft only.
Degree-2 Polynomial Regression	0.5627	■81.83 lakh	Curved area relationship adds very little.
Multiple Linear Regression	0.9178	■35.49 lakh	Best observed result among these three models.

Interpretation: Multiple Linear Regression performed best among the three compared models because it used several relevant features. An R^2 of about 0.9178 means the model explains about 91.78% of target variation on this test set; it does not mean every prediction is 91.78% accurate. An RMSE of ■35.49 lakh describes the scale of prediction error and is sensitive to large errors.

Business use: The model could support preliminary valuation and pricing analysis. It should be validated on new property data and reviewed for unusually large errors before being used for real financial decisions.

7. Final Submission Checklist

• Save the final notebook (.ipynb) with all code cells executed successfully.

• Include a model evaluation table with MSE, MAE, RMSE, R^2, and adjusted R^2 where calculated.

• Save relevant plots: correlation heatmap, linear-vs-polynomial regression plot, actual-vs-predicted plot if available, and Gradient Descent loss comparison.

• Record the actual training times and convergence results for Batch GD, SGD, and Mini-Batch GD.

• Write the final conclusion and mention that the best model is selected based on held-out test results.

• Check that dataset paths work and do not claim plots or files are included unless they are actually saved.

Quick revision: Regression predicts numbers; MAE is average absolute error; MSE squares errors; RMSE is in the target's units; R^2 summarizes explained variation; Gradient Descent minimizes loss; compare train and test performance to assess generalization.